

CS 513 Final Project



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What is our Project About?

- Predicting depression using a dataset found at <https://www.kaggle.com/datasets/hopesb/student-depression-dataset>.
- The dataset contains 27,898 student survey responses with 20+ features, including sleep habits, social behavior, motivation, and academic performance.
- We use machine learning to identify patterns that may indicate depressive symptoms based on students' emotional and behavioral data.
- Our goal is to support early detection and intervention efforts by building a model that flags students potentially at risk.

Student Depression Dataset

id	Gender	Age	City	Profession	Academic Pressure	Work Pressure	CGPA	Study Satisfaction	Job Satisfaction	Sleep Duration	Dietary Habits
2	Male	33.0	Visakhapatnam	Student	5.0	0.0	8.97	2.0	0.0	5-6 hours	Healthy
8	Female	24.0	Bangalore	Student	2.0	0.0	5.9	5.0	0.0	5-6 hours	Moderate
26	Male	31.0	Srinagar	Student	3.0	0.0	7.03	5.0	0.0	Less than 5 hours	Healthy
30	Female	28.0	Varanasi	Student	3.0	0.0	5.59	2.0	0.0	7-8 hours	Moderate
32	Female	25.0	Jaipur	Student	4.0	0.0	8.13	3.0	0.0	5-6 hours	Moderate
33	Male	29.0	Pune	Student	2.0	0.0	5.7	3.0	0.0	Less than 5 hours	Healthy
52	Male	30.0	Thane	Student	3.0	0.0	9.54	4.0	0.0	7-8 hours	Healthy
56	Female	30.0	Chennai	Student	2.0	0.0	8.04	4.0	0.0	Less than 5 hours	Unhealthy
59	Male	28.0	Nagpur	Student	3.0	0.0	9.79	1.0	0.0	7-8 hours	Moderate
62	Male	31.0	Nashik	Student	2.0	0.0	8.38	3.0	0.0	Less than 5 hours	Moderate
83	Male	24.0	Nagpur	Student	3.0	0.0	6.1	3.0	0.0	5-6 hours	Moderate
91	Male	33.0	Vadodara	Student	3.0	0.0	7.03	4.0	0.0	Less than 5 hours	Healthy
94	Male	27.0	Kalyan	Student	5.0	0.0	7.04	1.0	0.0	Less than 5 hours	Moderate
100	Female	19.0	Rajkot	Student	2.0	0.0	8.52	4.0	0.0	Less than 5 hours	Unhealthy
103	Female	19.0	Kalyan	Student	5.0	0.0	5.64	5.0	0.0	Less than 5 hours	Moderate
106	Male	29.0	Srinagar	Student	3.0	0.0	8.58	3.0	0.0	More than 8 hours	Moderate
120	Male	25.0	Nashik	Student	5.0	0.0	6.51	2.0	0.0	Less than 5 hours	Unhealthy

Degree	Have you ever had suicidal thoughts ?	Work/Study Hours	Financial Stress	Family History of Mental Illness	Depression
B.Pharm	Yes	3.0	1.0	No	1
BSc	No	3.0	2.0	Yes	0
BA	No	9.0	1.0	Yes	0
BCA	Yes	4.0	5.0	Yes	1
M.Tech	Yes	1.0	1.0	No	0
PhD	No	4.0	1.0	No	0
BSc	No	1.0	2.0	No	0
Class 12	No	0.0	1.0	Yes	0
B.Ed	Yes	12.0	3.0	No	1
LLB	Yes	2.0	5.0	No	1
Class 12	Yes	11.0	1.0	Yes	1
BE	Yes	10.0	2.0	Yes	0
M.Tech	No	10.0	1.0	Yes	1
Class 12	No	6.0	2.0	Yes	0
Class 12	Yes	4.0	5.0	Yes	1
M.Tech	Yes	10.0	2.0	Yes	1
M.Ed	Yes	2.0	5.0	Yes	1

Data Cleaning Strategy

- Any rows with null values are dropped
- “id” column dropped, as it is not necessary for classification
- Categorical columns with binary responses (“Yes”/“No”) mapped to 1 or 0
- Ordinal data (“Sleep Duration”, “Dietary Habits”) are given values in [1, n]
 - Rows with value of “Others” are dropped, only accounts for ~12 of 28k rows
- Purely categorical data is one-hot encoded in the preprocessor step
- Very few non-student participants, which are dropped
 - “Profession”, “Work Pressure”, & “Job Satisfaction” also dropped, as they are the same values for all students

Data Cleaning Strategy - Code Samples

```
df = df.drop("id", axis=1) # drop id (not needed for training)
```

```
# Convert these categorical yes/no questions to 1/0
```

```
df["Have you ever had suicidal thoughts ?"] = df["Have you ever had suicidal thoughts ?"].map(lambda x: 1 if x == "Yes" else 0)
```

```
df["Family History of Mental Illness"] = df["Family History of Mental Illness"].map(lambda x: 1 if x == "Yes" else 0)
```

```
df = df[df["Sleep Duration"] != "Others"] # drop "Others" from Sleep Duration
```

```
df = df[df["Dietary Habits"] != "Others"] # drop "Others" from Dietary Habits
```

```
# Convert Sleep Duration to numerical values
```

```
# 1 => Less than 5 hours
```

```
# 2 => 5-6 hours
```

```
# 3 => 7-8 hours
```

```
# 4 => More than 8 hours
```

```
df["Sleep Duration"] = df["Sleep Duration"].map(lambda x: 1 if x == "Less than 5 hours" else 2 if x == "5-6 hours" else 3 if x == "7-8 hours" else 4)
```

```
# Convert Dietary Habits to numerical values
```

```
# 1 => Unhealthy
```

```
# 2 => Moderate
```

```
# 3 => Healthy
```

```
df["Dietary Habits"] = df["Dietary Habits"].map(lambda x: 1 if x == "Unhealthy" else 2 if x == "Moderate" else 3)
```

Data Cleaning Strategy - Cleaned Data Shape & Values

	Gender	Age	City	Academic Pressure	CGPA	Study Satisfaction	Sleep Duration	Dietary Habits	Degree	Have you ever had suicidal thoughts ?	Work/Study Hours	Financial Stress	Family History of Mental Illness	Depression
0	Male	33.0	Visakhapatnam	5.0	8.97	2.0	2	3	B.Pharm	1	3.0	1.0	0	1
1	Female	24.0	Bangalore	2.0	5.90	5.0	2	2	BSc	0	3.0	2.0	1	0
2	Male	31.0	Srinagar	3.0	7.03	5.0	1	3	BA	0	9.0	1.0	1	0
3	Female	28.0	Varanasi	3.0	5.59	2.0	3	2	BCA	1	4.0	5.0	1	1
4	Female	25.0	Jaipur	4.0	8.13	3.0	2	2	M.Tech	1	1.0	1.0	0	0

```
Index: 27837 entries, 0 to 27900
Data columns (total 14 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Gender                                27837 non-null  object
1   Age                                   27837 non-null  float64
2   City                                  27837 non-null  object
3   Academic Pressure                     27837 non-null  float64
4   CGPA                                  27837 non-null  float64
5   Study Satisfaction                    27837 non-null  float64
6   Sleep Duration                        27837 non-null  int64
7   Dietary Habits                        27837 non-null  int64
8   Degree                                27837 non-null  object
9   Have you ever had suicidal thoughts ? 27837 non-null  int64
10  Work/Study Hours                      27837 non-null  float64
11  Financial Stress                      27837 non-null  float64
12  Family History of Mental Illness      27837 non-null  int64
13  Depression                            27837 non-null  int64
dtypes: float64(6), int64(5), object(3)
```

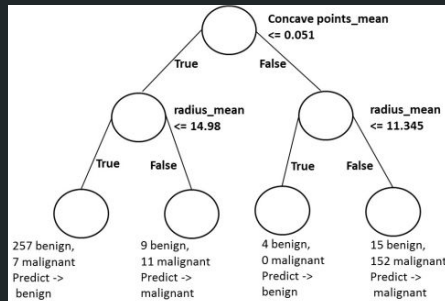
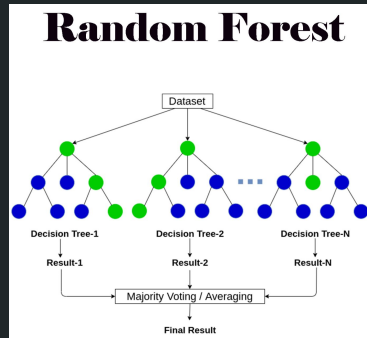
Summary Statistics

	Age	Academic Pressure	CGPA	Study Satisfaction	Sleep Duration	Dietary Habits	Have you ever had suicidal thoughts ?	Work/Study Hours	Financial Stress	Family History of Mental Illness	Depression
count	27837.000000	27837.000000	27837.000000	27837.000000	27837.000000	27837.000000	27837.000000	27837.000000	27837.000000	27837.000000	27837.000000
mean	25.819772	3.140748	7.656175	2.943385	2.398929	1.904228	0.632611	7.158889	3.139814	0.483996	0.585192
std	4.906603	1.381791	1.470791	1.361073	1.126706	0.796932	0.482102	3.706690	1.437217	0.499753	0.492698
min	18.000000	0.000000	0.000000	0.000000	1.000000	1.000000	0.000000	0.000000	1.000000	0.000000	0.000000
25%	21.000000	2.000000	6.290000	2.000000	1.000000	1.000000	0.000000	4.000000	2.000000	0.000000	0.000000
50%	25.000000	3.000000	7.770000	3.000000	2.000000	2.000000	1.000000	8.000000	3.000000	0.000000	1.000000
75%	30.000000	4.000000	8.920000	4.000000	3.000000	3.000000	1.000000	10.000000	4.000000	1.000000	1.000000
max	59.000000	5.000000	10.000000	5.000000	4.000000	3.000000	1.000000	12.000000	5.000000	1.000000	1.000000

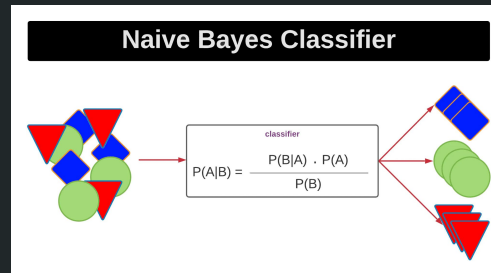
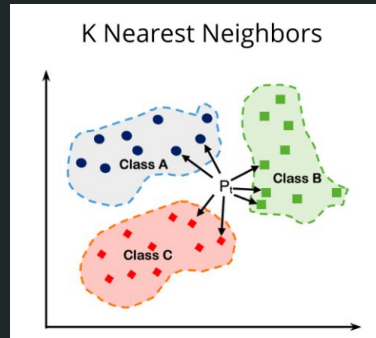
What Models did we Use?

We used 6 models, 2 for each member of our group.

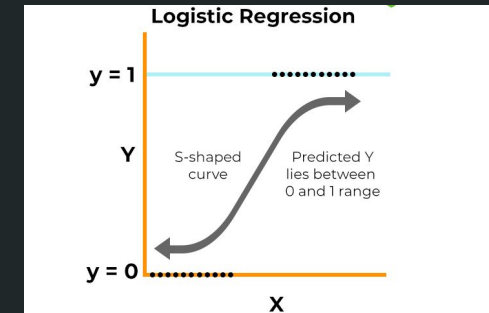
Random Forest and CART - Suprith



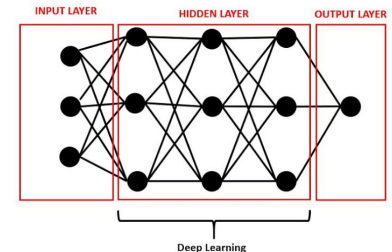
KNN and Naive Bayes - Ramses



Logistic Regression, ANN, SVM - John



ANN (Artificial Neural Network)

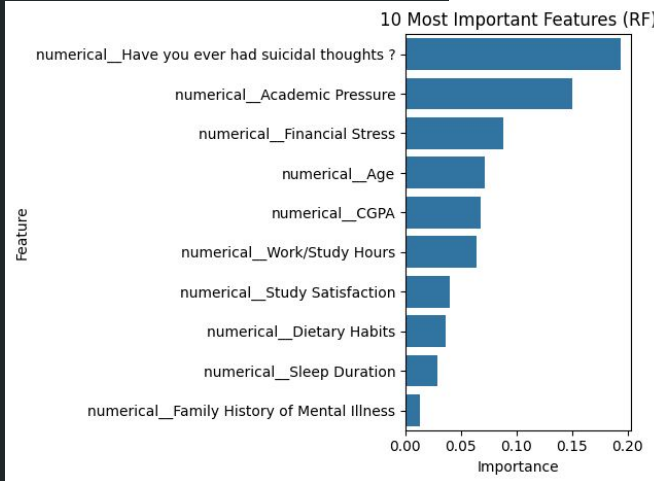
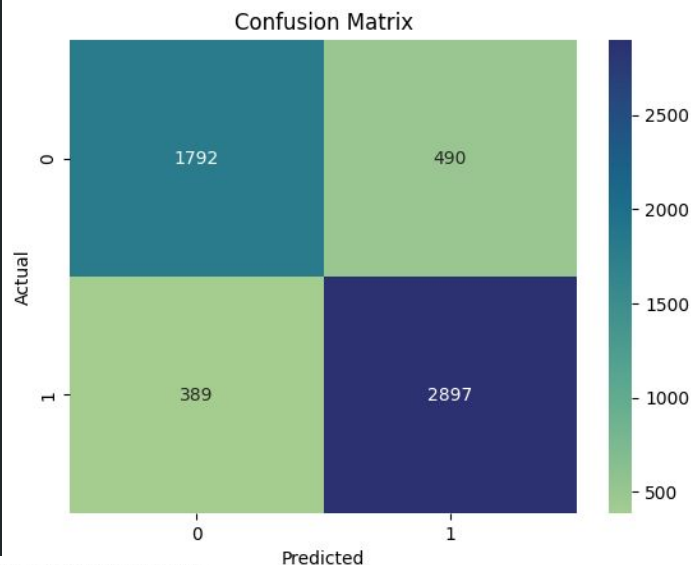


Random Forest Classification - Suprith

```
[[1792  490]
 [ 389 2897]]
```

	precision	recall	f1-score	support
0	0.82	0.79	0.80	2282
1	0.86	0.88	0.87	3286
accuracy			0.84	5568
macro avg	0.84	0.83	0.84	5568
weighted avg	0.84	0.84	0.84	5568

Random Forest accuracy: 0.8421336206896551



CART Classification - Suprith

```
[[1695  587]
 [ 646 2640]]
```

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

0	0.72	0.74	0.73	2282
---	------	------	------	------

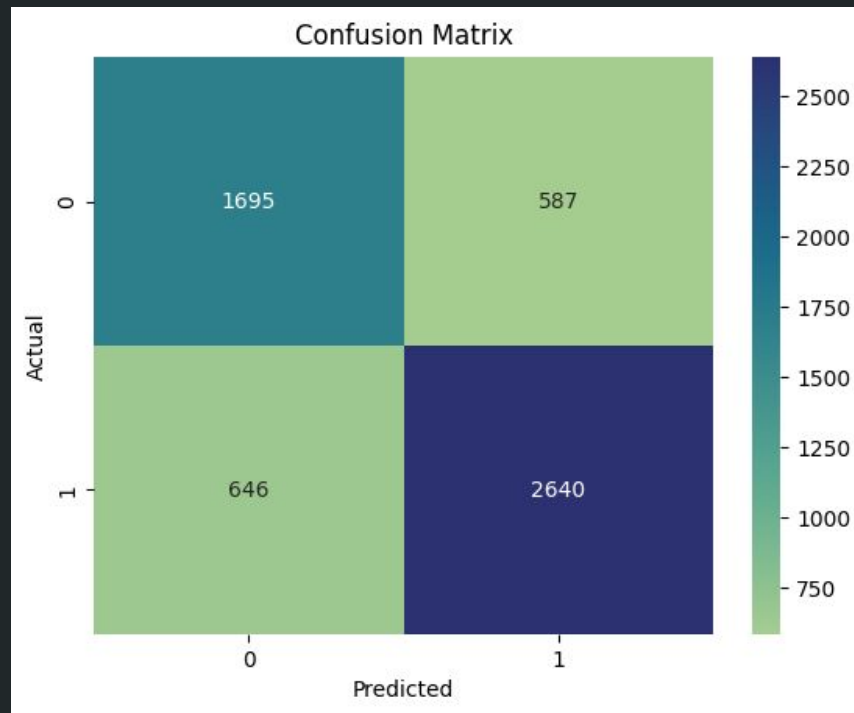
1	0.82	0.80	0.81	3286
---	------	------	------	------

accuracy			0.78	5568
----------	--	--	------	------

macro avg	0.77	0.77	0.77	5568
-----------	------	------	------	------

weighted avg	0.78	0.78	0.78	5568
--------------	------	------	------	------

CART accuracy: 0.7785560344827587



KNN Classification - Ramses

```
[[1490  792]
 [ 525 2761]]
```

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

0	0.74	0.65	0.69	2282
---	------	------	------	------

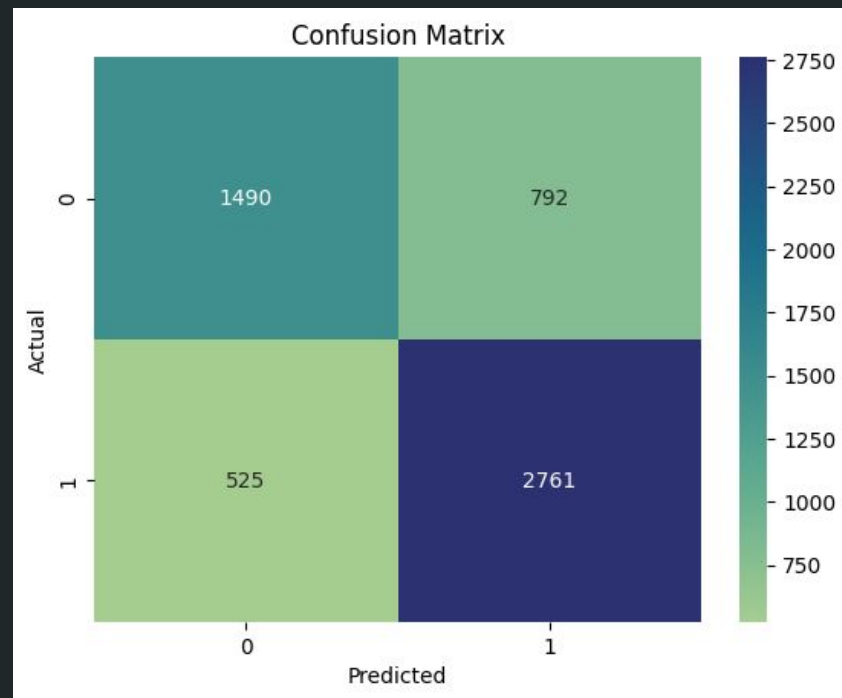
1	0.78	0.84	0.81	3286
---	------	------	------	------

accuracy			0.76	5568
----------	--	--	------	------

macro avg	0.76	0.75	0.75	5568
-----------	------	------	------	------

weighted avg	0.76	0.76	0.76	5568
--------------	------	------	------	------

KNN accuracy: 0.7634698275862069

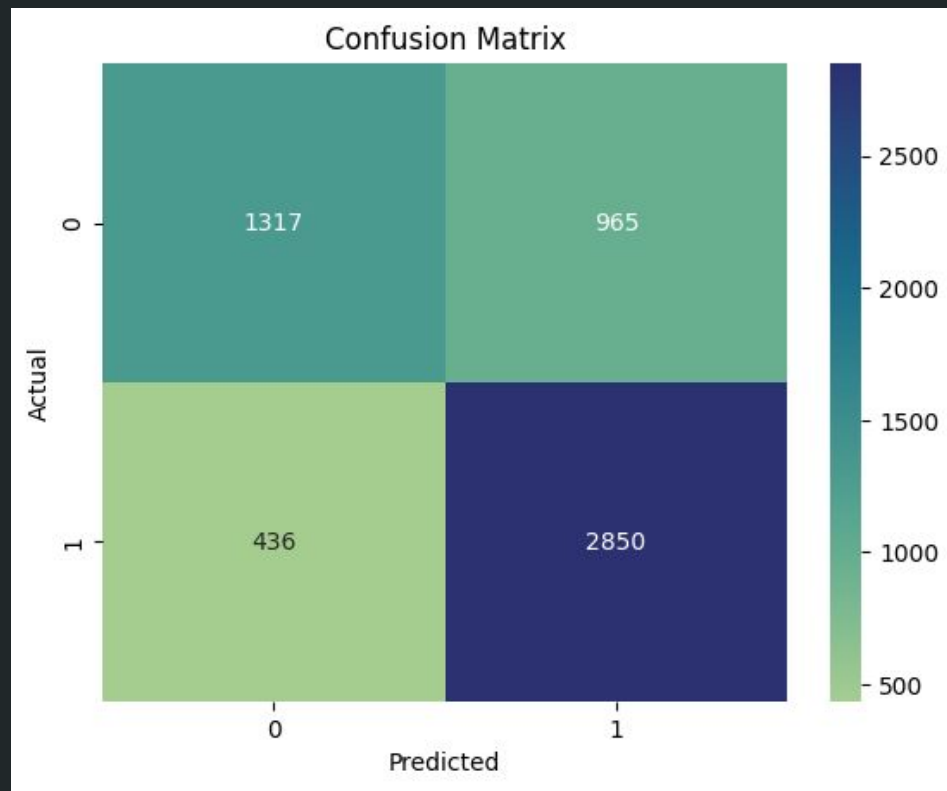


Naive Bayes Classification - Ramses

```
[[1317  965]
 [ 436 2850]]
```

	precision	recall	f1-score	support
0	0.75	0.58	0.65	2282
1	0.75	0.87	0.80	3286
accuracy			0.75	5568
macro avg	0.75	0.72	0.73	5568
weighted avg	0.75	0.75	0.74	5568

Naive Bayes accuracy: 0.7483836206896551

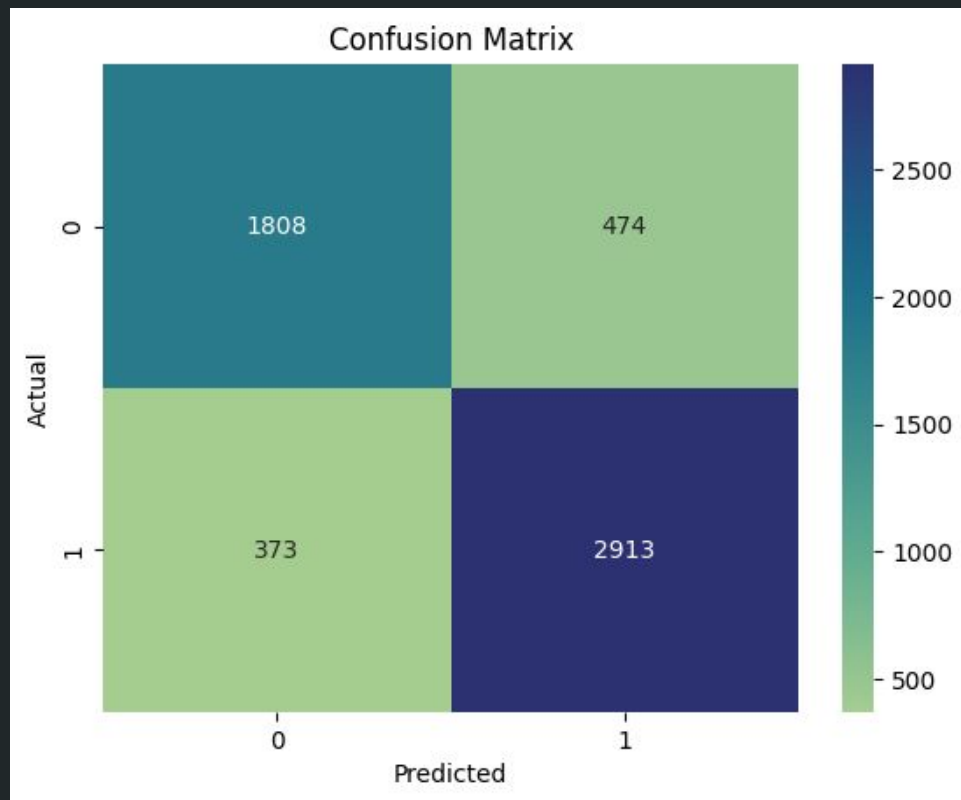


Logistic Regression Classification - John

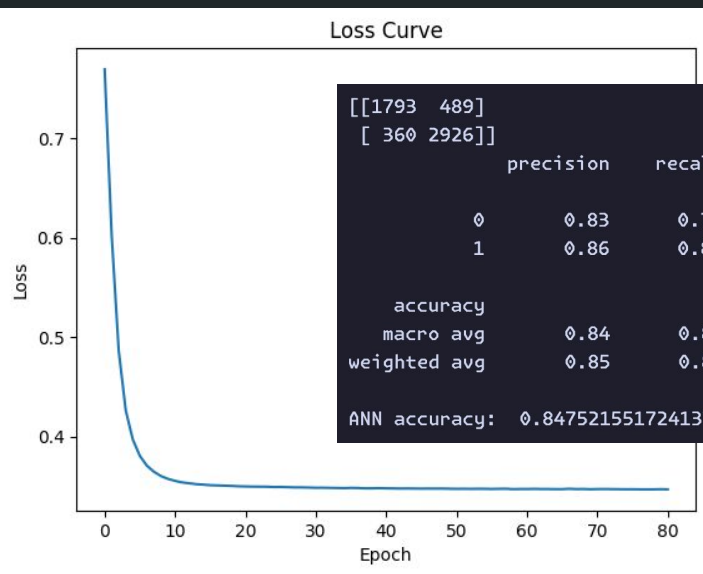
```
[[1808  474]
 [ 373 2913]]
```

	precision	recall	f1-score	support
0	0.83	0.79	0.81	2282
1	0.86	0.89	0.87	3286
accuracy			0.85	5568
macro avg	0.84	0.84	0.84	5568
weighted avg	0.85	0.85	0.85	5568

Logistic Regression accuracy: 0.8478807471264368



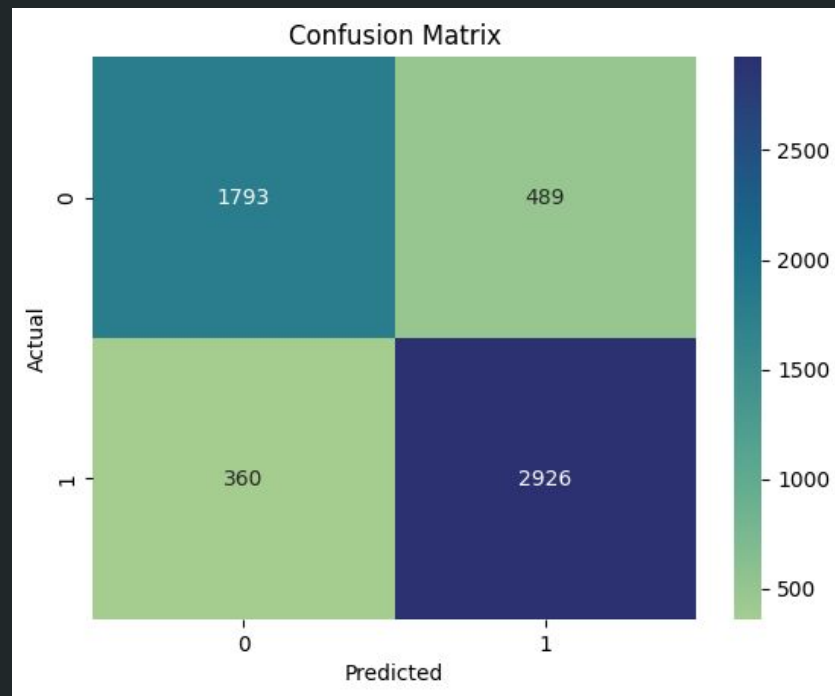
ANN Classification - John



```
[[1793 489]
 [ 360 2926]]
```

	precision	recall	f1-score	support
0	0.83	0.79	0.81	2282
1	0.86	0.89	0.87	3286
accuracy			0.85	5568
macro avg	0.84	0.84	0.84	5568
weighted avg	0.85	0.85	0.85	5568

ANN accuracy: 0.8475215517241379



```
model = Pipeline(steps=[
    ('preprocessor', nb_preprocessor),
    ('classifier', MLPClassifier(hidden_layer_sizes=(4,), max_iter=10000, random_state=RANDOM_STATE))
])
```

ANN Classification - Comparison of Parameters

- Among the models studied in class, ANN is one of the most customizable
- Previous results used default(-ish) parameters
- Substituted different activation/solver functions, learning rates, and hidden layer sizes, and compared resulting accuracies
 - Also altered normalization, using StandardScaler instead of MinMaxScaler
- Very slight, marginal improvements, accuracy remains around 84%

```
ACTIVATION = ['identity', 'logistic', 'tanh']  
SOLVER = ['lbfgs', 'sgd', 'adam']  
ALPHA = [0.0001, 0.01, 0.1]  
HIDDEN_LAYER_SIZES = [(2,), (8,), (16,)]
```

Best Results:

```
param (activation) = tanh, acc = 0.8489583333333334  
param (alpha) = 0.1, acc = 0.8489583333333334  
param (activation) = logistic, acc = 0.8482399425287356  
param (solver) = lbfgs, acc = 0.8477011494252874  
param (solver) = adam, acc = 0.8475215517241379
```

Support Vector Machine Classification - John

```
[[1798  484]
 [ 388 2898]]
```

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

0	0.82	0.79	0.80	2282
---	------	------	------	------

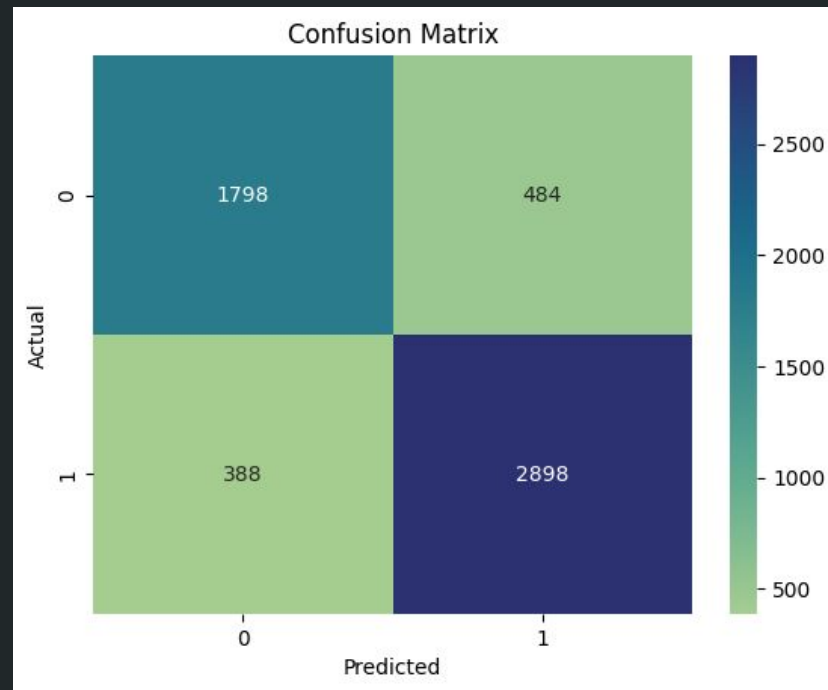
1	0.86	0.88	0.87	3286
---	------	------	------	------

accuracy			0.84	5568
----------	--	--	------	------

macro avg	0.84	0.83	0.84	5568
-----------	------	------	------	------

weighted avg	0.84	0.84	0.84	5568
--------------	------	------	------	------

SVM accuracy: 0.8433908045977011



Findings of Interest

- Naive Bayes, KNN have lowest accuracy, ~75%
 - Best performing models are Logistic Regression, ANN, ~85%
 - Tuning parameters of ANN does not show any impact for this dataset
-
- Removal of non-student-related columns drastically improves Naive Bayes performance
 - Accuracy jumped from 40% to the current 75%

Possible Limitations

- Average & median age of 25, 3rd quartile of 30, max age of 59
 - Possible difference between starting college at 18 vs. later in life
 - Adults are more financially stable
 - Likely continuing education as a choice, rather than other factors
- Imprecise survey answers, specifically for “Sleep Duration” and “Dietary Habits”
 - Broad categories instead of specific numerical ranges
 - Feature importance: these columns are less important than purely numerical columns
- Lack of other life factors, biological measurements, etc.
 - Previously diagnosed, comorbidities, psychiatric medications
 - Hormone levels (thyroid), nutrients, and vitamins
 - Sexual orientation, home life

The End

Thank you!